MILITARY SPECIFICATION

GRADERS, ROAD, PACKAGING OF

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the minimum requirements for methods, materials, and procedures for the preservation, packing, and marking of road graders and their attachments.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATION:

FEDERAL

L-P-378	- Plastic Sheet and Strip, Thin Gauge, Polyolefin.
QQ-S-781	- Strapping, Steel and Seals.
TT-P-664	- Primer Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting.
UU-T-81	- Tags, Shipping and Stock.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiber Board.
PPP-B-1055	- Barrier Material, Waterproofed, Flexible.
PPP-P-40	- Packaging and Packing of Hand Tools.

* U.S. GOVERNMENT PRINTING OFFICE: 1981—703-023/2086

FSC PACK

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-P-116	- Preservation-Packaging, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-R-196	- Repair Parts, Accessories, and Kits, Mechanical, Packaging of.
MIL-L-2105	- Lubricating Oil, Gear, Multi-purpose.
MIL-C-5501	- Cap and Plug, Protective, Dust and Moisture Seal.
MIL-E-10062	- Engines: Preparation for Shipment and Storage of.
MIL-T-22085	- Tape, Pressure-sensitive, Adhesive, Preservation and Sealing.
MIL-B-22191	- Barrier Material, Transparent, Flexible, Heat Sealable.
MIL-P-46093	- Primer Coating, Synthetic (For Brake Drums).
MIL-A-46153	- Antifreeze, Ethylene Glycol, Inhibited, Heavy Duty, Single Package.
MIL-B-46176	- Brake Fluid Silicone, Automotive All Weather, Operational and Preservative.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids, or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606.)

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NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Traffic Department, 1616 P. Street NW, Washington, D. C. 20036)

3. REQUIREMENTS

3.1 Materials. Materials shall be as specified herein and in applicable specifications and standards. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification. Materials shall be free of defects (see 6.5).

3.2 Preproduction pack. Unless otherwise specified (see 6.2) the contractor shall furnish a preproduction pack for examination within the time frame specified (see 6.2) to prove, prior to starting production packaging, that the applied preservation, packing, and marking comply with the requirements of this specification. Examination shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.3). The preproduction pack may be accomplished utilizing either the preproduction grader or a production grader. If the preproduction model grader is utilized, any preservation and packing shall be removed by the contractor at no expense to the Government, when requested by the Government to facilitate comparison between the preproduction grader and the production grader.

3.3 Preservation. Preservation shall be level A, level B, or commercial, as specified (see 6.2).

3.3.1 Level A.

3.3.1.1 Disassembly. Disassembly shall be the minimum necessary to safe-guard parts vulnerable to damage, pilferage, and loss. The extent of further disassembly shall be confined to the removal of protruding parts that would otherwise increase cubage; however, disassembly shall be limited to those parts that can be removed and reinstalled without special tools. Removed bolts, nuts, pins, screws, and washers shall be reinstalled in mating parts and secured to prevent their loss. Keys removed shall be secured to the keyway of the primary component or shaft. The extent of further disassembly shall be confined to that required by the transportability requirements of the end-item specification.

3.3.1.2 Matchmarking. Parts removed and mating parts on the graders and attachments shall be matchmarked when necessary to facilitate reassembly. Removed parts and mating parts shall be identified by cloth shipping tags conforming to UU-T-81, type A, style optional, and the marked tags attached to mating parts. The information on the cloth tags shall be applied with water-proof ink.

3.3.1.3 Preservatives. Preservatives specified herein shall conform to the applicable specifications listed in and shall be applied in accordance with MIL-P-116.

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3.3.1.4 Unprotected surfaces. Unprotected exterior metal surfaces requiring the application of a contact preservative in accordance with MIL-P-116 and not specifically specified herein or in the end item specification shall be coated as follows.

3.3.1.4.1 Unfinished (not machined) surfaces. Unfinished exterior metal surfaces shall be coated with type P-1 preservative.

3.3.1.4.2 Machined surfaces. Exposed machined surfaces shall be coated with type P-6 or P-11 preservative and wrapped or covered, as applicable with barrier material conforming to MIL-B-121, type I or II, grade A, class 2 or MIL-B-22191, type II. The material shall be secured in place with tape conforming to MIL-T-22085, type II.

3.3.1.5 Engines. Engines, engine components, and engine accessories shall be preserved in accordance with the level A requirements of MIL-E-10062, type II, method I.

3.3.1.5.1 Cooling systems. Engines with liquid cooling systems shall be preserved by the water and antifreeze procedure specified in MIL-E-10062, except the antifreeze shall conform to MIL-A-46153.

3.3.1.6 Instrument panels. Instruments, meters, and gages not protected by a metal cover or by the grader cab shall have the instrument glass or gage glass covered with barrier material conforming to MIL-B-121, type II, grade A, class 2 over which a fitted piece of hardboard or 1/4-inch-thick plywood shall be placed and secured with tape conforming to MIL-T-22085, type II.

3.3.1.7 Exposed gears. Exposed gears shall be coated with type P-1 preservative or with primer conforming to TT-P-664 or MIL-P-46093.

3.3.1.8 Enclosed gears. Gears operating on lubricating oil (SAE 10, 30, or 50) shall have the housing filled to the operating level with type P-10 preservative, type I, grade 10, 30, or 50, as applicable and tagged to indicate: "The preservative oil is adequate for operation. Do not drain until the first required lubrication change." Gears operating on gear lubricant (SAE 80 or 90) shall be filled to the operating level with lubricant conforming to MIL-L-2105, grade as applicable, and tagged to indicate: "The lubricant is good for operation. Do not drain until the first required lubrication change." Gears not operating on lubricating oil (SAE 10, 30, or 50) or lubricant (SAE 80 or 90) shall be filled to the operating level with the approved lubricant required for operation. The gear housings shall be tagged to indicate: "The housing is filled to the operating level with the applicable type lubricant. Do not drain prior to the first required lubrication change." The tags shall conform to UU-T-81, type A. The information on the tags shall be applied with waterproof ink.

3.3.1.9 Automatic transmissions and torque converters. The housings shall be filled to the operating level with the lubricant required for operation. The transmissions and torque converters shall be operated at sufficient engine speed to insure coating of all interior surfaces and parts with the lubricant.

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Openings into the housing shall be sealed with tape conforming to MIL-T-22085, type II. A tag shall be attached to the control lever indicating: "The housing is filled to the operating level with the approved lubricant and is good for operation. Do not operate engine until the tape has been removed from the openings." The tag shall conform to UU-T-81, type A. The information on the tag shall be applied with waterproof ink.

3.3.1.10 Disc-type clutches. With the cover plates removed and the clutch engaged, all accessible metal interior components of the clutch shall be sprayed with a thin film of primer conforming to TT-P-664 or primer conforming to MIL-P-46093. When specified (see 6.2), the clutch shall be disassembled sufficiently to permit the application of a continuous thin film (max. 0.75 mil) of the same primer to friction surfaces of the pressure plates. Clutch control mechanisms not enclosed shall be coated with type P-1 preservative. After being sprayed, spring-loaded-type clutches shall have the clutch pedal secured in a partially disengaged position to eliminate contact between the disc facings and pressure plates. Snap-over-center and toggle-in-type clutches shall be completely disengaged. Cover plates shall be reinstalled.

3.3.1.11 Exposed drive chains. Exposed drive chains shall be coated with enough type P-3 or P-9 preservative to insure penetration of the preservative to the inner surfaces of the rollers, pins, and bushings. After the excess preservative has drained, the entire chain and the unpainted surfaces of the sprocket shall be coated with type P-1 preservative.

3.3.1.12 Enclosed chains. Enclosed chains and chain housings shall be preserved and tagged as specified in 3.3.1.8 for enclosed gears.

3.3.1.13 Drive belts and pulleys. Drive belts shall be removed or released from tension. Removed belts shall be preserved in accordance with MIL-P-116, method IC-1 or IC-3. Unpainted faces of pulley grooves shall be coated with primer conforming to TT-P-664 or primer conforming to MIL-P-46093. A tag conforming to UU-T-81, type A, shall be attached in a conspicuous location indicating: "Belts have been removed or released from tension. Install or tension as applicable, prior to operation". The information on the tag shall be applied with waterproof ink.

3.3.1.14 Drive shafts and universal joints. Unpainted surfaces of drive shafts and universal joints shall be coated with type P-1 preservative. Dust boots, if furnished, shall be reinstalled.

3.3.1.15 Flywheel and starter drives. When specified (see 6.2), flywheel ring gears and starter drive mechanisms shall be coated with type P-10 preservative, type I, grade 30.

3.3.1.16 Winterization system. Interior surfaces of the tanks, pumps, piping and burners requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with type P-10 preservative, type I, grade 30. The threaded surfaces of fill caps and drain plugs shall be coated with the preservative and reinstalled.

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3.3.1.17 Hydraulic systems (except hydraulic brakes or steering systems). The hydraulic fluid supply tanks shall be filled to the operating level with hydraulic fluid required for operation. The pistons shall be retracted as far as practicable into the cylinders and secured. When the pistons cannot be fully retracted, the exposed portions of the piston rods (ramshafts) shall be coated with type P-6 or P-11 preservative and the coated surfaces wrapped or covered with barrier-material conforming to MIL-B-121, type II, grade A, class 2, extending the wraps approximately 2 inches on the ram cylinders. The wraps shall be secured in place with tape conforming to MIL-T-22085, type II. When the pistons can be fully retracted, any remaining uncoated surfaces of the piston rods shall be coated with type P-1 preservative, with no wrapping required. The hydraulic control valves shall be secured in the neutral position and preserved as specified herein for piston rods. Hoses shall not be disconnected. Any unpainted exterior metal surfaces of the system, requiring the application of a contact preservative in accordance with MIL-P-116, shall be coated with type P-1 preservative. A tag conforming to UU-T-81, type A, shall be attached to the control lever indicating: "The hydraulic supply tank is filled to the operating level with fluid required for operation. Do not drain." Waterproof ink shall be used for printing the information on the tag.

3.3.1.18 Brakes.

3.3.1.18.1 General. Exterior, unpainted, untreated, and threaded surfaces of cables, clevises, and linkages of service and parking brakes, requiring the application of a contact preservative in accordance with MIL-P-116, shall be coated with type P-1 preservative. When specified (see 6.2), metal surfaces enclosed within brake drums (such as adjusting wedges, pins, eccentrics, cam levers, linkages, anchors, retracting springs) and the braking surfaces of the brake drums shall be sprayed with a continuous film of primer conforming to TT-P-664 or primer conforming to MIL-P-46093. The film shall be a minimum of 0.50 mil to a maximum of 0.75 mil thick as determined by an elcometer. Primer material containing lumps or skin, that will not return to a homogenous mixture upon stirring, shall not be used. The film shall be thoroughly dry prior to reinstalling the brake drums. A warning tag conforming to UU-T-81, type A, shall be securely attached in a conspicuous location in the operator's compartment, indicating "Brake drums and components preserved. Do not apply brakes when grader is being moved. Towing shall be accomplished by use of a rigid towbar or similar arrangement until this grader is issued to the user." Waterproof ink shall be used for the information marking on the tag. Note: Interior surfaces of brake drums shall not be preserved when the grader design incorporates a planetary drive with the wheel brake assembly and the surfaces are inaccessible by simply removing the brake drums.

3.3.1.18.2 Hydraulic brakes and steering systems.

3.3.1.18.2.1 Operating on nonpetroleum base hydraulic fluid. The complete hydraulic system of brakes and steering systems that operate on nonetroleum base hydraulic fluids shall be filled to the operating level with presservative fluid conforming to MIL-B-46176 and the system actuated to insure coating of all interior surfaces with the preservative. A tag conforming to UU-T-81, type A, shall be attached to the controls indicating: "This system is filled with

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preservative fluid and is good for indefinite storage and for operation in ambient temperatures not lower than minus 10° F. Prior to operation in ambient temperatures lower than minus 10° F, the system shall be drained and refilled to the operating level with the hydraulic fluid required for operation." Waterproof ink shall be used for printing the information on the tag.

3.3.1.18.2.2 Operating on petroleum base hydraulic fluid. The complete hydraulic system of brakes and steering systems that operate on petroleum base fluid shall be filled to the operating level with the fluid required for operation and the systems actuated to insure coating of all interior surfaces with the fluid. A tag conforming to UU-T-81, type A, shall be attached to the system controls indicating: "This system is filled with the approved operating fluid." Waterproof ink shall be used for printing the information on the tag.

3.3.1.18.3 Air-actuated brake systems. The air-actuated brake systems shall be drained. Interior surfaces of the air-supply tanks shall be fogged with type P-10 preservative, type I, grade 30. Excess preservative oil shall be drained. Threaded openings and the threads of drainplugs shall be coated with the preservative oil specified herein. The drainplugs shall be reinstalled. Air-line filters shall be drained and closed. The exhaust ports of relay emergency, quick-release, and relay valves not equipped with exhaust check valves shall be closed by inserting caps or plugs as applicable, conforming to MIL-C-5501 or by sealing the ports with tape conforming to MIL-T-22085, type II. Tags conforming to UU-T-81, type A, shall be attached to the valves and service lines indicating: "The plugs, caps, or tape must be removed from the exhaust ports and valves before this grader is placed in operation." Waterproof ink shall be used for printing the information on the tags.

3.3.1.19 Electrical components. Openings in sockets, coupling plugs, and switches shall be sealed with tape conforming to MIL-T-22085, type II.

3.3.1.20 Lamps, reflectors, and mirrors. Lamps, reflectors, and mirrors, if removed, shall be individually preserved in accordance with MIL-P-116, method III, using boxes conforming to PPP-B-636, W5c or W6c, style optional. Lamps, reflectors and mirrors not removed shall be covered with barrier material conforming to MIL-B-121, type II, Grade A, class 2. The barrier material shall be secured in place with tape conforming to MIL-T-22085, type II.

3.3.1.21 Gear shifts and control levers. Unpainted metal surfaces of handles and levers shall be coated with type P-1 preservative. Composition and rubber handgrips, when not protected by a full cab, shall be removed if practical and preserved in accordance with MIL-P-116, Method IC-1 or IC-3. When the handgrips are left in place they shall be wrapped with barrier material conforming to PPP-B-1055, Class E-2, or black polyethylene conforming to L-P-378 (6 mil) and the wrap secured with tape conforming to MIL-T-22085, type II.

3.3.1.22 Seat backs, arm rests, and cushions. Seat backs, arm rests and cushions when not protected by a full cab, shall be removed if practical and individually wrapped with barrier material conforming to PPP-B-1055, class E-2 or black polyethylene conforming to L-P-378 (6 mil), and the wrap secured with tape conforming to MIL-T-22085, type II. When seat backs, arm rests, and

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cushions are left in place they shall be wrapped or covered with the barrier material specified herein and the wrap or cover secured in place with tape specified herein.

3.3.1.23 Windshield wiper arms and blades. Windshield wiper arms and blades shall be removed and preserved in accordance with MIL-P-116, method IC-1 or IC-3. The preserved wiper arms and blades shall be placed in the toolbox. The exposed wiper shaft shall be coated with type P-1 preservative.

3.3.1.24 Extra cutting edges and end bits. Unpainted surfaces of extra cutting edges and end bits shall be coated with type P-1 preservative. The extra cutting edges shall be secured to the grader. (see 3.4.1.1).

3.3.1.25 Locks, keys, and key openings. Interiors of locks shall be coated with powdered graphite. The locks shall be operated to insure penetration of the graphite. Openings into locks (except padlocks) shall be sealed with tape conforming to MIL-T-22085, type II. Switch keys, cab door keys, and padlocks with keys shall be preserved in accordance with MIL-P-116, Method IC-1 or IC-3. The preserved keys and padlocks with keys for each grader shall be placed in the toolbox. A tag shall be attached in a conspicuous location in the operator's compartment indicating: "The keys for the grader are in the toolbox." For graders without cabs the information on the tags shall be applied with a waterproof ink.

3.3.1.26 Rubber tires. Tires mounted on the wheels (road tires) of the grader shall be inflated to 10 pounds above pressure value recommended for maximum load. Tires mounted on rims (spare tires) shall be inflated to two-thirds of the specified operating pressure.

3.3.1.27 Fire extinguishers. Unpainted exterior metal surfaces of fire extinguishers requiring the application of a contact preservative in accordance with MIL-P-116, shall be coated with type P-1 preservative. Fire extinguishers enclosed with full cabs shall be secured in their mounting brackets. Fire extinguishers not enclosed within a full cab shall be individually placed in a box conforming to PPP-B-636, Class Weather-Resistant. Blocking and cushioning shall be provided within the box to prevent movement or damage. Boxes shall be sealed in accordance with the appendix to the box specification, method V.

3.3.1.28 Maintenance tools. Maintenance tools shall be preserved in accordance with the level A preservation and packaging requirements of PPP-P-40.

3.3.1.29 Technical publications. Technical publications for each grader shall be preserved together in accordance with MIL-P-116, method IC-1 or IC-3.

3.3.1.30 Repair parts. Unless otherwise specified, the preservative application criteria and applicable method(s) of preservation of MIL-P-116 shall be used to preserve repair parts. When specified (see 6.2), the repair parts shall be preserved in accordance with the level A requirements of MIL-R-196, or when parts are not specifically covered in MIL-R-196, an applicable sub-method of preservation of MIL-P-116 shall be used.

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3.3.1.31 Other components. Other components not specifically mentioned herein requiring protection from corrosion or physical or mechanical damage shall be preserved as specified for components of similar design and construction.

3.3.1.32 Consolidation.

3.3.1.32.1 Components. Components for each grader, except extra cutting edges, shall be consolidated together in boxes conforming to PPP-B-621, class 2, style optional; or PPP-B-601, overseas type, style optional. The boxes shall be close-fitting. The boxes shall be strapped in accordance with the appendix to the applicable box specification.

3.3.1.32.2 Repair parts. Repair parts for each grader or for attachments not attached to the grader shall be consolidated together in boxes as specified in 3.3.1.32.1.

3.3.1.32.3 Maintenance tools and technical publications. Maintenance tools and technical publications for each grader shall be placed in the toolbox.

3.3.1.33 Toolbox. The toolbox lid shall be closed and secured to prevent pilferage.

3.3.1.34 Attachments. Attachments not attached to the grader, such as snowplows, rollers, scarifiers, backhoes, bulldozer blade, loaders, and scrapers, shall have the hydraulic systems, gears, chains, and other components preserved as specified herein for similar components of attachments installed on the grader. Hydraulic lines and other openings into the attachments shall be sealed by dummy couplers if so equipped, or with caps or plugs conforming to MIL-C-5501, or with tape conforming to MIL-T-22085, type II. Consolidation of disassembled components for the attachment shall be in boxes specified in 3.3.1.32.1. The consolidated containers shall be secured to the attachment for which intended with flat steel strapping conforming to QQ-S-781, type I or IV, regular duty, finish B, size as applicable, in a manner not to interfere with lifting or moving the attachment.

3.3.2 Level B.

3.3.2.1 Disassembly. Disassembly shall be as specified in 3.3.1.1 for level A.

3.3.2.2 Matchmarking. Matchmarking shall be as specified in 3.3.1.2 for level A.

3.3.2.3 Preservatives. Preservatives specified herein shall conform to the applicable specifications listed in and shall be applied in accordance with MIL-P-116.

3.3.2.4 Unprotected surfaces. Unpainted exterior metal surfaces requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with preservative as specified in 3.3.1.4.

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3.3.2.5 Engines. Engines and engine accessories shall be preserved as specified in 3.3.1.5 for level A.

3.3.2.6 Instrument panels. Instruments and gage glasses not protected by a metal cover or by the grader cab shall be protected as specified in 3.3.1.6 for level A.

3.3.2.7 Enclosed gears and chains. The housings shall be filled to the operating level with the applicable type lubricant required by the lubricating order for operation and tagged to indicate that the housings are filled with the required amount of operating oil.

3.3.2.8 Automatic transmissions and torque converters. The housings shall be filled to the operating level with the applicable type lubricant required by the lubricating order for operation and tagged to indicate that the housings are filled with the required amount of operating oil.

3.3.2.9 Disc-type clutches. Spring-loaded type clutches shall have the control secured in a partially disengaged position. Snap-over-center and toggle-in type clutches shall be completely disengaged.

3.3.2.10 Exposed drive chains. Exposed drive chains shall be coated with type P-3 or P-9 preservative.

3.3.2.11 Drivebelts. Drivebelts shall be released from tension and tagged as indicated in 3.3.1.13 for level A.

3.3.2.12 Winterization system. The winterization systems shall be preserved as specified in 3.3.1.16 for level A.

3.3.2.13 Hydraulic systems (including hydraulic brakes and steering). The hydraulic fluid supply tanks or reservoirs shall be filled with the hydraulic fluid required for operation and the systems shall be tagged to indicate the supply tanks or reservoirs are filled with the hydraulic fluid required for operation.

3.3.2.14 Air-actuated brake systems. Air-actuated brake systems shall be preserved as specified in 3.3.1.18.3 for level A.

3.3.2.15 Lamps, reflectors, and mirrors. Lamps, reflectors, and mirrors if removed from the grader shall be preserved as specified in 3.3.1.20 for level A.

3.3.2.16 Seat backs, arm rests, and cushions. Seat backs, arm rests and cushions shall be protected as specified in 3.3.1.22 for level A.

3.3.2.17 Windshield wiper arms and blades. Windshield wiper arms and blades shall be removed and placed in the toolbox.

3.3.2.18 Rubber tires. Tires shall be inflated as specified in 3.3.1.26 for level A.

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3.3.2.19 Fire extinguishers. Fire extinguishers enclosed within the cab shall be secured in the mounting brackets. Fire extinguishers not enclosed within the cab shall be individually placed in a close-fitting box conforming to PPP-B-636, class domestic variety and grade as applicable to the weight involved.

3.3.2.20 Maintenance tools, technical publications, and repair parts. Maintenance tools, technical publications, and repair parts shall be preserved as specified for level A.

3.3.2.21 Consolidation. Consolidation shall be as specified for level A, except boxes shall be domestic type or class.

3.3.2.22 Tool box. The tool box lid shall be closed and secured to prevent pilferage.

3.3.3 Commercial. Each complete grader, attachments, repair parts, maintenance tools, and technical publications (except for engine and engine accessories [see 3.3.3.1]) shall be preserved in accordance with MIL-STD-1188.

3.3.3.1 Engine. Liquid cooling systems shall be filled with a fresh clean solution of 50 percent water and 50 percent antifreeze conforming to MIL-A-46153. A tag shall be attached indicating the temperature to which the cooling system can be subjected to before damage. All lubrication reservoirs, sumps and hydraulic fluid supply tanks or reservoirs shall be filled to the operating level with the lubricant or fluid designated for use in the temperature range at the shipping destination. A tag or tags, as applicable shall be attached in a conspicuous location indicating which lubricant or fluid has been used.

3.3.3.2 Tool box. The tool box lid shall be closed and secured to prevent pilferage.

3.4 Packing. Packing shall be level A, level B, or commercial as specified (see 6.2).

3.4.1 Level A. The graders preserved as specified in 3.3, shall be prepared for mobile (unboxed) shipment as follows.

3.4.1.1 Components and consolidated containers. Preserved and unpreserved components, consolidated containers, and extra cutting edges shall be secured on the grader with strapping conforming to QQ-S-781, class 1, type I or IV, finish B, size as applicable. The items shall be secured in such a manner that the minimum cube will be realized, wheels shall be free to rotate, and arrangement shall not interfere with towing or lifting the grader with slings.

3.4.1.2 Moldboard and circle. The circle shall be centered under the main grader frame and the moldboard shall be turned so that the ends do not extend beyond the outside of the tires and wheels. The circle and moldboard shall be raised to the maximum height and locked in position, and for hydraulically controlled graders shall be additionally secured to prevent movement by tying to the grader frame with wire rope, size as applicable, secured with wire rope clips or with strapping conforming to QQ-S-781, class 1, type I or IV, finish B, size as applicable.

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3.4.1.3 Scarifier. The scarifier shall be raised to its maximum height and locked, and for hydraulically controlled graders shall be additionally secured by tying to the grader frame with wire rope, size as applicable, secured with wire rope clips or with strapping conforming to QQ-S-781, class 1, type I or IV, finish B, size as applicable.

3.4.1.4 Leaning-wheel graders. Leaning wheels shall be positioned vertically and locked in place.

3.4.1.5 Engine housings. The housing doors shall be closed and secured. Louvers and other small openings shall be sealed with tape conforming to MIL-T-22085, type II. Engines not completely enclosed within housings shall be covered with barrier material conforming to PPP-B-1055, class E-2 secured with tape as specified herein. When specified (see 6.2), radiator fronts shall be covered with barrier material specified herein and secured with tape as specified herein.

3.4.1.6 Cabs. Cab windows and doors shall be closed. When specified (see 6.2), windshields and glass windows shall be protected from breakage by covering each glass with a fitted piece of hardboard, or with 1/4-inch-thick plywood or a sheet metal panel secured in place with bolts or with steel strapping conforming to QQ-S-781, class I, type I or IV, finish B, size as applicable. There shall be not less than 1 inch clearance between the panels and the glass. The panels shall be painted the same color as the cab. Small openings into the cab shall be sealed with tape conforming to MIL-T-22085, type II.

3.4.1.7 Attachments. Attachments not attached to the grader shall be shipped unboxed. Controls shall be locked or secured in the neutral position. Loose ends of hoses shall be secured to the attachment in a manner to prevent damage. The ends of hoses shall be sealed with plastic or metal caps or plugs as applicable.

3.4.2 Level B. The graders, components, repair parts, tools and publications preserved as specified in 3.3 shall be prepared for mobile (unboxed) shipment as specified for level A, except as follows: Strapping shall not be required to be finish B. Sealing of openings into engine housings and shrouding of engines shall not be required, except where overseas shipment is involved. Attachments not attached to grader shall be shipped unboxed as specified in 3.4.1.7 for level A.

3.4.3 Commercial. The graders, components, attachments, repair parts, tools, and publications preserved as specified in 3.3 shall be prepared for mobile (unboxed) shipment in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules.

3.5 Marking.

3.5.1 Military. Marking for military levels of protection (level A or B) shall be in accordance with MIL-STD-129.

3.5.2 Commercial. Marking for commercial packaging shall be in accordance with MIL-STD-1188.

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3.6 Depreservation guides. Two depreservation guides shall be prepared and placed in waterproof envelopes marked "Depreservation Guide." One of the guides shall be secured to the steering wheel or steering controls of the grader. The other copy shall be placed in the grader toolbox. Unless otherwise specified (see 6.2), "DA Form 2258 - Depreservation Guide for Vehicles and Equipment" shall be used.

3.7 Workmanship. Workmanship shall be such as to provide maximum protection to the graders by application of the requirements specified herein, to prevent corrosion, deterioration, and mechanical or physical damage during handling, shipment, and storage.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Material inspection. The contractor is responsible for insuring that supplies and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspections. Inspections shall be classified as follows:

(a) Preproduction pack inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction pack inspection.

4.3.1 Examination. The preproduction pack shall be examined for the defects specified in 4.4.4. Examination shall be in three stages as specified in 4.4.1. Presence of one or more defects shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Inspection stages.

(a) The first stage shall include inspection of procedures, containers, materials, methods, and marking prior to enclosing the loose components in consolidated containers.
(b) The second stage shall include inspection of the marking and closure of the consolidated containers.

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(c) The third stage shall include inspection of the procedures methods, materials, and marking of the complete mobile (unboxed) grader as prepared for shipment. This stage shall include arrangement of consolidated containers and components required to be secured to the grader for shipment.

4.4.2 Unit of product. For the purpose of inspection a complete unit prepared for any applicable inspection stage shall be considered a unit of product.

4.4.3 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.4.4 Examination. Samples selected in accordance with 4.4.3 shall be examined for the defects marked "X" for the applicable level in table I. Acceptable Quality Level (AQL) shall be 2.5 percent defective.

TABLE I. Defects.

MAJOR	DEFECTS	LEVEL		
		A	B	COMMERCIAL
101.	Materials, methods, and containers, not as specified. Each incorrect material, each incorrect method, and each incorrect container shall constitute one defect, (3.1, 3.3, 3.4).	X	X	X
102.	Disassembly not as specified, (3.3.1.1, 3.3.2.1).	X	X	-
103.	Matchmarking not as specified, (3.3.1.2, 3.3.2.2).	X	X	-
104.	Information on tags not applied with waterproof ink, (3.3.1.2, 3.3.1.8, 3.3.1.9, 3.3.1.13, 3.3.1.17, 3.3.1.18.1, 3.3.1.18.2.1, 3.3.18.2.2, 3.3.1.18.3, 3.3.1.25).	X	X	-
105.	Unprotected surfaces not coated with preservative as specified (3.3.1.4, 3.3.2.4).	X	X	-
106.	Coated surfaces not wrapped or covered as specified, (3.3.1.4, 3.3.2.4).	X	X	-
107.	Engines and engine components not preserved as specfied (3.3.1.5, 3.3.2.5, 3.3.3.1).	X	X	X
108.	Instrument panels not protected as specified (3.3.1.6, 3.3.2.6).	X	X	-
109.	Exposed gears not coated with preservative or with primer as specified (3.3.1.7).	X	-	-

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TABLE I. Defects. (Cont'd)

MAJOR	DEFECTS	LEVEL		
		A	B	COMMERCIAL
110.	Enclosed gear housing not filled with preservative oils or with operational lubricant as specified and the housings not tagged as specified (3.3.1.8, 3.3.2.7)	X	X	-
111.	Automatic transmissions and torque converters not filled with operational lubricant and the control levers not tagged as specified (3.3.1.9, 3.3.2.8)	X	X	-
112.	Accessible interior metal surfaces of clutches not coated with primer as specified (3.3.1.10)	X	-	-
113.	Clutches not completely disengaged or completely disengaged as applicable (3.3.1.10, 3.3.2.9)	X	X	-
114.	Exposed drive chains not coated with preservative as specified (3.3.1.11, 3.3.2.10).	X	X	-
115.	Enclosed chain housings not filled with preservative oil or not filled with operational lubricant as applicable and the housings not tagged as specified (3.3.1.12, 3.3.2.7).	X	X	-
116.	Drive belts not removed or released from tension as specified and not tagged as specified (3.3.1.13, 3.3.2.11).	X	X	-
117.	Unpainted surfaces of pulley grooves not coated with primer as specified (3.3.1.13).	X	-	-
118.	Drive shafts and universal joints not coated with preservative as specified and dust boots not reinstalled as specified (3.3.1.14).	X	-	-
119.	Winterization system components not preserved as specified (3.3.1.16, 3.3.2.12).	X	X	-
120.	Hydraulic systems not filled with fluid as specified (3.3.1.17, 3.3.2.13).	X	X	-
121.	Exposed portions of hydraulic piston rods not preserved as specified and the preserved surfaces not wrapped when required (3.3.1.17).	X	-	-
122.	Hydraulic systems not tagged as specified (3.3.1.17, 3.3.2.13).	X	X	-

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TABLE I. Defects. (Cont'd)

MAJOR	DEFECTS	LEVEL A	LEVEL B	LEVEL COMMERCIAL
123.	Surfaces of brake components enclosed within brake drums and the braking surfaces of brake drums not coated with primer when specified (3.3.1.18.1).	X	-	-
124.	Exterior surfaces of brakes not coated with preservative as specified (3.3.1.18.1).	X	-	-
125.	Tag not attached to indicate brakes have been preserved (3.3.1.18.1, 3.3.1.18.2.1, 3.3.1.18.2.2, 3.3.2.13).	X	X	-
126.	Hydraulic brake systems operating on nonpetroleum base fluid not filled with preservative fluid as specified (3.3.1.18.2.1, 3.3.2.13).	X	X	-
127.	Hydraulic brake systems operating on petroleum base fluid not filled to the operating level with the fluid required for operation (3.3.1.18.2.2, 3.3.2.13).	X	X	-
128.	Air-actuated brake system supply tank not fogged with preservative oil as specified, excess preservative not drained, ports and valves not sealed and tagged as specified (3.3.1.18.3, 3.3.2.14).	X	X	-
129.	Openings into electrical sockets, coupling plugs, and switches not sealed as specified (3.3.1.19).	X	-	-
130.	Lamps, reflectors, and rear vision mirrors not preserved or not covered as applicable (3.3.1.20, 3.3.2.15).	X	X	-
131.	Composition or rubber handgrips not preserved or not wrapped as applicable (3.3.1.21).	X	-	-
132.	Seat backs, armrests, and cushions not wrapped or not covered as specified (3.3.1.22, 3.3.2.16).	X	X	-
133.	Windshield wiper arms and blades not removed, preserved and placed in tool box as specified (3.3.1.23, 3.3.2.17).	X	X	-
134.	Interior of locks not coated with graphite and openings into locks not sealed as specified (3.3.1.25).	X	-	-

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TABLE I. Defects. (Cont'd)

MAJOR	DEFECTS	LEVEL A	LEVEL B	LEVEL COMMERCIAL
135.	Key and padlocks not placed in toolbox as specified (3.3.1.25).	X	-	-
136.	Tag not attached within operators compartment to indicate location of keys (3.3.1.25).	X	-	-
137.	Rubber tires not inflated as specified (3.3.1.26, 3.3.2.18).	X	X	-
138.	Unpainted exterior surfaces of fire extinguishers not coated with preservative, if required (3.3.1.27).	X	-	-
139.	Fire extinguishers not placed in box as specified, if required (3.3.1.27) or secured in mounting brackets (3.3.2.19).	X	X	-
140.	Maintenance tools not preserved in accordance with the referenced document (3.3.1.28, 3.3.2.20, 3.3.3).	X	X	X
141.	Technical publications not preserved as specified (3.3.1.29, 3.3.2.20, 3.3.3).	X	X	X
142.	Repair parts not preserved in accordance with the referenced document (3.3.1.30, 3.3.2.20, 3.3.3).	X	X	X
143.	Consolidation not as specified (3.3.1.32, 3.3.2.21.	X	X	-
144.	Toolbox lids not secured to prevent pilferage (3.3.1.33, 3.3.2.22, 3.3.3.2).	X	X	X
145.	Consolidated containers not secured on the grader in the manner specified (3.4.1.1, 3.4.2).	X	X	-
146.	Strapping not as specified (3.4.1.1, 3.4.1.6, 3.4.2).	X	X	-
147	Moldboard and circle not positioned and secured as specified (3.4.1.2, 3.4.2).	X	X	-
148.	Scarifiers not positioned, not locked, and not secured as specified (3.4.1.3, 3.4.2).	X	X	-
149.	Leaning wheels not positioned as specified (3.4.1.4, 3.4.2).	X	X	-

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TABLE I. Defects. (Cont'd)

MAJOR	DEFECTS	LEVEL A	LEVEL B	LEVEL COMMERCIAL
150.	Engines not protected by a shroud when not fully enclosed (3.4.1.5).	X	-	-
151.	Radiator fronts not covered as specified, when required (3.4.1.5).	X	-	-
152.	Windshields and window glass not protected as specified, when required (3.4.1.6, 3.4.2).	X	X	-
153.	Marking missing, illegible, incomplete, incorrect (3.5).	X	X	X
154.	Depreservation guides not prepared and stowed as specified (3.6).	X	X	X

5. PACKAGING

(This section not applicable to this specification.)

6. NOTES

6.1 Intended use. It is intended that the requirements of this specification be used for reference in Section 5 of equipment specifications, for direct reference in contract or orders, and can be used as a guide in the preparation of packaging data sheets covering preservation, packing, and marking of roadgraders.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) When a preproduction pack is not required (see 3.2).
(c) Time frame required for submission of preproduction pack (see 3.2).
(d) Degree of preservation and degree of packing required (see 3.3 and 3.4).
(e) When friction surfaces of clutch pressure plates are to be preserved (see 3.3.1.10).
(f) When flywheel ring gears and starter drive mechanism are to be preserved (3.3.1.15).
(g) When brake components enclosed within brake drums and the braking surfaces of brake drums are to be preserved (see 3.3.1.18.1).
(h) Applicable document to be used for preserving repair parts (see 3.3.1.30)
(i) When radiator fronts are to be protected (see 3.4.1.5).
(j) When windshields and window glass is to be protected (see 3.4.1.6).
(k) When other than DA Form 2258 is to be used (see 3.6).

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6.3 Preproduction pack. Any changes or deviations of production packs from the approved preproduction pack will be subject to the approval of the contracting officer. Approval of the preproduction pack will not relieve the contractor of his obligation to preserve, pack, and mark the graders and components in accordance with this specification.

6.4 Depreservation guides. The contracting officer should arrange to furnish the necessary copies of DA Form 2258, when requested by the contractor (see 3.6).

6.5 Recycled materials. It is encouraged that recycled materials be used, when practical, as long as it meets the requirements of this specification (see 3.1).

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